242652: felsic metavolcanic rock, Bowgarder Hills

(Youanmi Terrane, Yilgarn Craton)

Location and sampling

Perenjori (SH 50-6), Perenjori (2139)
MGA Zone 50, 432328E 6765598N

Sampled on 4 May 2019

This sample was collected from outcrop adjacent to a power line access track on the west side of the Bowgarder Hills, about 13.5 km west of Weelhamby Lake, 10.7 km east-southeast of Koolanooka mine, and 4.5 km south of Mungada Road.

Geological context

The unit sampled is a felsic metavolcanic rock within the Koolanooka greenstone belt, which consists mainly of greenschist to amphibolite facies volcanic, sedimentary, and ultramafic to felsic intrusive rocks, and banded iron-formation (Baxter et al., 1983; Baxter and Lipple, 1985; Watkins, 1990). A metarhyolite collected in the southern Koolanooka greenstone belt, about 15.9 km to the south-southeast, yielded an igneous crystallization age of 3002 ± 4 Ma (GSWA 242672, Wingate et al., 2021). A metamonzogranite collected about 19.8 km to the northeast yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al., 2020).

Petrographic description

The sample is an altered, fine-grained, felsic metavolcanic rock (Fig. 1), consisting of about 75% sericite- and epidote-altered microcrystalline material, 5% larger quartz grains, 5–7% actinolitic hornblende, 5–7% epidote, 3% biotite, 3% chlorite, 2–3% carbonate minerals, and trace sulfide minerals, prehnite and zircon. The microcrystalline material is a quartz-rich granoblastic aggregate of mainly transparent grains with an average size of <0.025 mm. Extensive sericite and epidote alteration suggests that feldspar is present, but is either altered or not twinned. Larger quartz grains, up to 0.3 mm in size, are rounded or irregular, and disseminated in aggregates with secondary minerals. Although they may be primary in origin, they do not appear to be typical volcanic phenocrysts. The remaining minerals are secondary, and either metamorphic or low-temperature alteration products. Blue–green actinolitic hornblende is anhedral to subhedral, and up to 0.3 mm in size. Pale brown biotite is anhedral, up to 0.25 mm in size. Chlorite has partially replaced both biotite and hornblende. Epidote is finely disseminated throughout the microcrystalline material. Sulfide minerals include anhedral, irregular pyrrhotite up to 0.1 mm in size, and smaller grains of pyrite and chalcopyrite. Prehnite occurs in veins up to 0.1 mm wide. The sample has been metamorphosed to a hornblende hornfels, characterized by a finely granoblastic texture and an assemblage of actinolitic hornblende and biotite. Epidote, chlorite, carbonate and at least some quartz are lower-temperature secondary minerals. The rock lacks volcanic textures and may represent a siliceous tuffaceous rock rather than a coherent rhyolite.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, and subhedral to euhedral. The crystals are up to 200 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, although about half of the zircons luminesce brightly whereas the others are dark. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed during three analytical sessions, on 20–21 and 26–27 March and 14–15 May 2020, using SHRIMP-B. Analyses 1.1 to 12.1 (spot numbers 1–12) were obtained during the first session, together with 12 analyses of the M257 standard, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.20% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.42% (1σ). Analyses 13.1 to 24.1 (spot numbers 13–24) were obtained during the second session, together with 11 analyses of the M257 standard, of which eight analyses indicated an external spot-to-spot (reproducibility) uncertainty of 2.33% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.99% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the second session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.44%. Analyses 24.2 to 30.1 (spot numbers 25–36) were obtained during the third session, together with 10 analyses of the M257 standard which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.20% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the third session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.54%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Figure 1. **Hand specimen image of sample 242652: felsic metavolcanic rock, Bowgarder Hills**

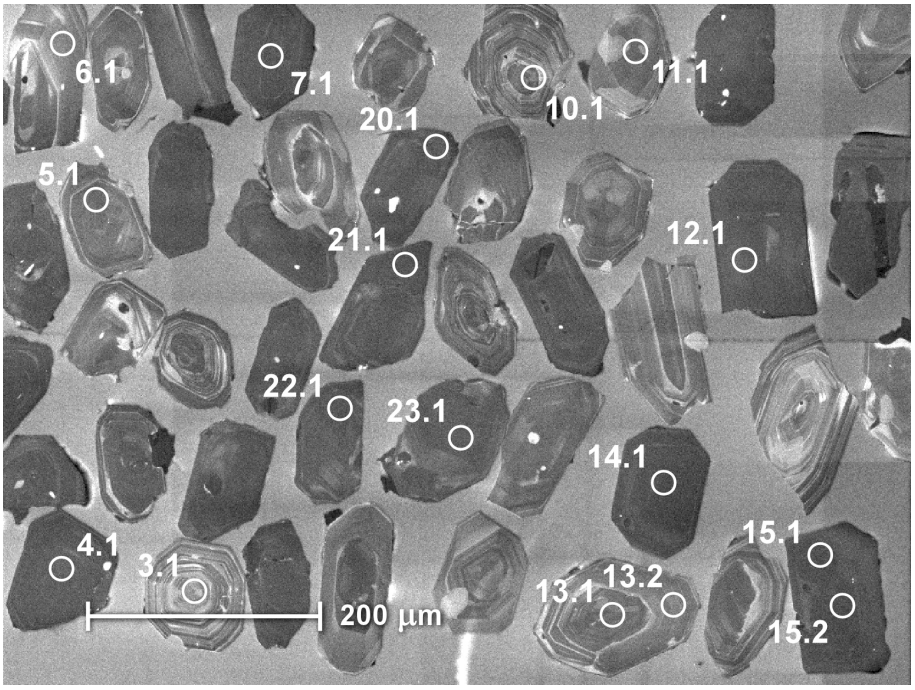

Figure 2. **Cathodoluminescence image of representative zircons from sample 242652: felsic metavolcanic rock, Bowgarder Hills. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 242652: felsic metavolcanic rock, Bowgarder Hills

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	32	13.2	133	36	0.27	0.038	1.772	0.019	0.21622	0.00148	1.773	0.019	0.21588	0.00131	2883	25	2950	10	2.3
I	26	25.1	179	84	0.47	0.027	1.761	0.017	0.21638	0.00086	1.762	0.017	0.21614	0.00168	2898	22	2952	13	1.8
I	36	30.1	173	66	0.38	0.029	1.793	0.018	0.21643	0.00085	1.794	0.018	0.21617	0.00120	2856	23	2952	9	3.3
I	19	19.1	147	56	0.38	0.080	1.707	0.046	0.21695	0.00155	1.709	0.046	0.21624	0.00158	2970	67	2953	12	-0.6
I	8	8.1	179	76	0.43	0.050	1.773	0.033	0.21692	0.00098	1.774	0.033	0.21648	0.00100	2882	44	2955	7	2.5
I	6	6.1	137	39	0.29	0.061	1.771	0.028	0.21725	0.00108	1.772	0.028	0.21671	0.00110	2885	38	2956	8	2.4
I	13	13.1	195	100	0.52	0.014	1.804	0.049	0.21716	0.00149	1.805	0.049	0.21703	0.00149	2843	64	2959	11	3.9
I	7	7.1	542	396	0.73	0.042	1.785	0.025	0.21744	0.00055	1.785	0.025	0.21707	0.00056	2867	33	2959	4	3.1
I	3	3.1	104	49	0.47	-0.055	1.814	0.040	0.21726	0.00132	1.813	0.040	0.21775	0.00135	2832	51	2964	10	4.5
I	27	26.1	181	80	0.44	0.030	1.705	0.017	0.21811	0.00128	1.705	0.017	0.21784	0.00122	2975	24	2965	9	-0.3
I	11	11.1	192	123	0.64	0.036	1.725	0.026	0.21837	0.00093	1.725	0.026	0.21805	0.00095	2947	37	2966	7	0.6
I	10	10.1	199	107	0.54	0.017	1.701	0.026	0.21844	0.00090	1.701	0.026	0.21829	0.00091	2980	37	2968	7	-0.4
I	5	5.1	208	101	0.49	0.032	1.775	0.027	0.21862	0.00088	1.776	0.027	0.21833	0.00089	2880	35	2968	7	3.0
I	9	9.1	168	81	0.48	0.050	1.760	0.027	0.21887	0.00099	1.760	0.027	0.21842	0.00101	2900	37	2969	7	2.3
X	20	20.1	404	191	0.47	0.040	1.798	0.050	0.21992	0.00123	1.799	0.050	0.21956	0.00124	2850	66	2977	9	4.3
X	28	18.2	327	146	0.44	0.018	1.709	0.014	0.22037	0.00092	1.709	0.014	0.22021	0.00105	2970	20	2982	8	0.4
X	33	27.1	276	115	0.42	0.064	1.739	0.015	0.22132	0.00111	1.740	0.015	0.22075	0.00108	2927	21	2986	8	2.0
X	25	24.2	546	328	0.60	0.047	1.699	0.013	0.22119	0.00109	1.700	0.013	0.22077	0.00090	2983	19	2986	7	0.1
X	22	22.1	324	242	0.75	0.007	1.682	0.050	0.22095	0.00123	1.682	0.050	0.22089	0.00123	3007	73	2987	9	-0.7
X	34	28.1	703	567	0.81	0.011	1.726	0.012	0.22140	0.00104	1.726	0.012	0.22130	0.00085	2946	17	2990	6	1.5
X	29	17.2	483	354	0.73	0.020	1.735	0.013	0.22148	0.00083	1.735	0.013	0.22130	0.00091	2934	18	2990	7	1.9
X	21	21.1	593	378	0.64	-0.012	1.715	0.046	0.22132	0.00112	1.715	0.046	0.22143	0.00112	2962	67	2991	8	1.0
X	15	15.1	583	336	0.58	-0.015	1.803	0.052	0.22156	0.00116	1.803	0.052	0.22169	0.00116	2845	69	2993	8	4.9
X	31	15.2	601	401	0.67	-0.012	1.762	0.013	0.22174	0.00116	1.762	0.013	0.22184	0.00089	2898	17	2994	6	3.2
X	14	14.1	559	371	0.66	0.004	1.687	0.055	0.22224	0.00185	1.687	0.055	0.22221	0.00185	3001	80	2997	13	-0.1
X	4	4.1	284	199	0.70	0.013	1.765	0.026	0.22237	0.00080	1.765	0.026	0.22226	0.00081	2893	35	2997	6	3.5
X	23	23.1	385	273	0.71	-0.018	1.646	0.043	0.22210	0.00121	1.645	0.043	0.22226	0.00121	3061	66	2997	9	-2.1
X	17	17.1	586	315	0.54	0.017	1.714	0.045	0.22248	0.00114	1.715	0.045	0.22232	0.00114	2962	64	2998	8	1.2
X	12	12.1	320	254	0.79	0.032	1.724	0.025	0.22274	0.00074	1.725	0.025	0.22246	0.00075	2948	35	2999	5	1.7
X	24	24.1	349	125	0.36	0.031	1.658	0.048	0.22279	0.00121	1.658	0.048	0.22252	0.00122	3042	73	2999	9	-1.4
X	2	2.1	627	419	0.67	0.041	1.780	0.024	0.22334	0.00056	1.780	0.024	0.22297	0.00056	2874	33	3002	4	4.3
P	1	1.1	64	26	0.40	0.051	1.688	0.031	0.21540	0.00155	1.689	0.031	0.21494	0.00159	2998	45	2943	12	-1.9
D	35	29.1	118	46	0.39	0.144	1.854	0.020	0.21980	0.00085	1.857	0.020	0.21853	0.00138	2778	25	2970	10	6.5
D	30	16.2	194	102	0.53	0.094	1.805	0.017	0.22424	0.00090	1.807	0.017	0.22342	0.00181	2840	22	3005	13	5.5
D	18	18.1	556	288	0.52	-0.009	2.388	0.066	0.22546	0.00133	2.387	0.066	0.22554	0.00134	2255	54	3021	10	25.3
D	16	16.1	377	191	0.51	0.115	3.596	0.101	0.23135	0.00263	3.600	0.101	0.23033	0.00273	1580	40	3054	19	48.3

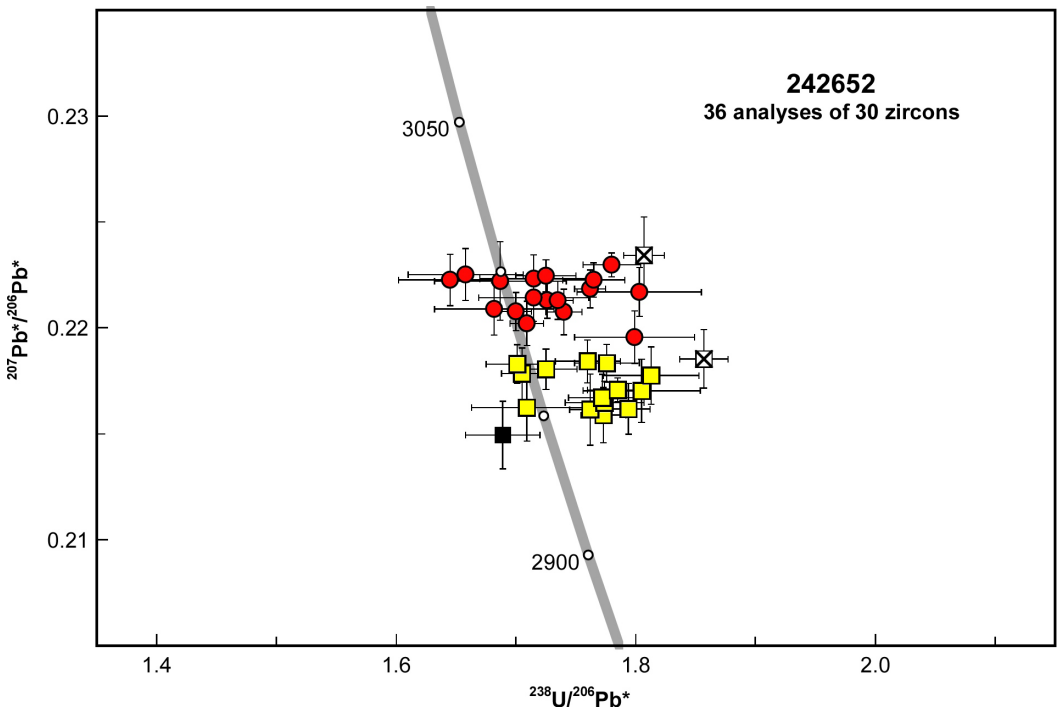

Figure 3. **U–Pb analytical data for sample 242652: felsic metavolcanic rock, Bowgarder Hills. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); black square indicates Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%). Two strongly discordant analyses in Group D are not shown**

Results

Thirty-six analyses were obtained from 30 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Four analyses are >5% discordant. The dates obtained from these four analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 32 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios, U and Th concentrations, and appearance in CL images.

Group I comprises 14 analyses of 13 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2961 ± 4 Ma (MSWD = 0.68). These analyses were located in zircons that are bright in CL images, and indicate mainly moderate concentrations of U (104–542 ppm, median 179 ppm) and Th (36–396 ppm, median 81 ppm).

Group X comprises 17 analyses of 14 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2994 ± 3 Ma (MSWD = 0.96). These analyses were located in zircons that are dark in CL images, and indicate mainly high concentrations of U (276–703 ppm, median 483 ppm) and Th (115–567 ppm, median 315 ppm).

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2943 ± 12 Ma (1σ). This analysis was located in a zircon that is bright in the CL image, and indicates low concentrations of U (64 ppm) and Th (26 ppm).

The date of 2961 ± 4 Ma for the 14 analyses in Group I is interpreted as the igneous crystallization age of the volcanic rock. The date of 2994 ± 3 Ma for the 17 analyses in Group X is interpreted as the age of an inherited component. The date of 2943 ± 12 Ma (1σ) for the single analysis in Group P is interpreted to reflect minor ancient loss of radiogenic Pb.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021, 242652: felsic metavolcanic rock, Bowgarder Hills; Geochronology Record 1807: Geological Survey of Western Australia, 5p.

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